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Antibiotic-Induced Depletion of Anti-inflammatory Clostridia Is Associated with the Development of Graft-versus-Host Disease in Pediatric Stem Cell Transplantation Patients

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A B S T R A C T

Adult stem cell transplantation (SCT) patients with graft-versus-host-disease (GVHD) exhibit significant disruptions in gut microbial communities. These changes are associated with higher overall mortality and appear to be driven by specific antibiotic therapies. It is unclear whether pediatric SCT patients who develop GVHD exhibit similar antibiotic-induced gut microbiota community changes. Here, we show that pediatric SCT patients (from Children's Medical Center Dallas, n = 8, and Cincinnati Children's Hospital, n = 7) who developed GVHD showed a significant decline, up to 10-log fold, in gut anti-inflammatory Clostridia (AIC) compared with those without GVHD. In fact, the development of GVHD is significantly associated with this AIC decline and with cumulative antibiotic exposure, particularly antibiotics effective against anaerobic bacteria (P = .003, Firth logistic regression analysis). Using metagenomic shotgun sequencing analysis, we were able to identify specific commensal bacterial species, including AIC, that were significantly depleted in GVHD patients. We then used a preclinical GVHD model to verify our clinical observations. Clindamycin depleted AIC and exacerbated GVHD in mice, whereas oral AIC supplementation increased gut AIC levels and mitigated GVHD in mice. Together, these data suggest that an antibiotic-induced AIC depletion in the gut microbiota is associated with the development of GVHD in pediatric SCT patients.

INTRODUCTION

The success of hematopoietic stem cell transplantation (SCT) is limited, in part, by graft-versus-host-disease (GVHD), a complex immune-mediated process occurring in 20% to 50% of HSCT patients [1] and accounting for 15% to 30% of deaths that occur after allogeneic SCT [2].

Commensal gut bacteria have long been implicated in initiating and perpetuating GVHD [3]. Recent studies using next-generation sequencing methods to profile gut microbiota have shown that adult SCT patients who develop GVHD have significantly less Clostridiales than counterparts who do not develop GVHD [4,5]. This group of nonpathogenic Clostridia (that we have previously termed *anti-inflammatory Clostridia* [AIC] [6], specifically including the Firmicutes families Clostridiaceae, Erysipelotrichaceae, Eubacteriaceae, Lachnospiraceae, and Ruminococcaceae) has been shown to ameliorate inflammatory bowel disease (IBD) [7,8] and GVHD [9] in mice. Of note, these gut microbiota changes have not been described in pediatric SCT patients who develop GVHD.

Antibiotic use appears to drive many of the microbiota changes observed in SCT patients with GVHD, including a loss of gut microbial diversity that is associated with increased

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overall mortality [5,10,11]. Yet given the extraordinarily high incidence of infection in SCT patients, prolonged antibiotic therapy during SCT is an unfortunate reality. Pediatric patients undergoing SCT may be even more prone to antibiotic-induced gut microbiota changes than their adult counterparts. First, pediatric patients (younger than 2 years of age) have immature gut microbiota and relatively fewer beneficial commensal anaerobes, including AIC [12,13]; thus, antibiotics may have a disproportionate effect on an immature gut microbiome. Second, antibiotics are the most commonly prescribed medications for children [14] and, thus, pretransplantation antibiotic exposure could affect gut microbiota baselines going into SCT. Thus, understanding how specific classes or types of antibiotics drive gut microbiota changes and how these affect the SCT patient's health (GVHD and infections) will be critical for making rational and safe antibiotic choices in the future.

Here, we have leveraged several methods for characterizing the gut microbiota of pediatric patients undergoing allogeneic SCT: bacterial group quantitative PCR (qPCR), 16S rRNA sequencing, and metagenomic shotgun sequencing (MSS). As shown in adult SCT patients, we observed a depletion of AIC in pediatric SCT patients who developed GVHD. Significant declines in AIC levels were noted in SCT patients who developed GVHD, and these changes were strongly associated with cumulative antibiotic exposure, particularly with antibiotics effective against anaerobes. We were also able to show that antibiotic-induced depletion of AIC exacerbated GVHD in mice and, conversely, that AIC supplementation ameliorated GVHD in mice.

MATERIALS AND METHODS

Patients

We designed a prospective case-cohort pilot study with the intent to characterize the gut microbiota of pediatric patients undergoing allogeneic SCT (alloSCT) with and without GVHD. All subjects who underwent transplantation at Children's Medical Center Dallas (Dallas cohort) were enrolled under the human study protocol 112010-110 through the institutional review boards of the University of Texas Southwestern Medical Center and Children's Medical Center Dallas. SCT patients with documented infections or pre-existing diseases of the gastrointestinal tract were excluded from enrollment. Thirty patients were enrolled between November 2011 and July 2013. One patient did not undergo transplantation and was excluded from the study. Of those, 10 patients had sufficient quantity and quality of fecal gDNA collected for full microbiome analysis, which included 16S rRNA, bacterial qPCR, and MSS. All patients received myeloablative conditioning, immune prophylaxis, and prophylactic antibiotics (Table 1). *Engraftment* after SCT was defined as an absolute neutrophil count ≥ 500/mm^3 for 3 consecutive days. Patients categorized as having GVHD were those patients with either biopsy-confirmed intestinal GVHD or high-stage (2 to 4) skin or liver GVHD. Because disruptions in gut microbiota communities have been associated with several extraintestinal autoimmune diseases [15], we hypothesized that significant changes in gut microbiota populations might also be associated with the development of high-stage skin or liver GVHD.

Table 1
Clinical Characteristics of Pediatric allo-SCT Patients Who Underwent Transplantation at Children's Medical Center Dallas (n = 8) and Cincinnati Children's Hospital (n = 7)

Characteristic	Combined	Dallas Cohort	Cincinnati Cohort
Date of transplantation	February 2012-August 2013	February 2012-November 2012	May 2013-August 2013
Age, median (range), yr	10.03 (1.6-14.5)	6.15 (1.6-14.5)	10.79 (5.06-14.46)
Gender	Female, 8 (53%); male, 7 (47%)	Female, 2 (25%); male, 6 (75%)	Female, 6 (86%); male, 1 (14%)
Primary diagnosis	SCA, 4 (27%); AML, 3 (20%); ALL, 2 (13%); HLH, 2 (13%); DBA, 1 (7%); FA, 1 (7%); SAA, 1 (7%); SCID, 1 (7%)	SCA, 3 (38%); AML, 2 (25%); DBA, 1 (13%); FA, 1 (13%); SCID, 1 (7%)	ALL, 2 (29%); HLH, 2 (29%); SCA, 1 (14%); AML, 1 (14%); SAA, 1 (14%)
Graft source	Bone marrow, 14 (93%); cord blood, 1 (7%); peripheral blood, 0	Bone marrow, 7 (88%); cord blood, 1 (13%); peripheral blood, 0	Bone marrow, 7 (100%); cord blood, 0; peripheral blood, 0
Conditioning intensity	SIM, 9 (60%); RIM, 6 (40%); NM, 0	SIM, 6 (75%); RIM, 2 (25%); NM, 0	SIM, 3 (43%); RIM, 4 (57%); NM, 0
HLA matching	HLA identical sibling, 7 (47%); HLA identical unrelated, 4 (27%); HLA mismatched unrelated, 4 (27%)	HLA identical sibling, 4 (50%); HLA identical unrelated, 2 (25%); HLA mismatched unrelated, 2 (25%)	HLA identical sibling, 3 (43%); HLA identical unrelated, 2 (29%); HLA mismatched unrelated, 2 (29%)
GVHD prophylaxis	Tacrolimus, 8 (53%); cyclosporine, 7 (47%); methotrexate, 6 (40%); mycophenolate, 8 (53%); prednisone, 4 (27%)	Tacrolimus, 8 (100%); cyclosporine, 0; methotrexate, 1 (13%); mycophenolate, 8 (100%); prednisone, 0	Tacrolimus, 0; cyclosporine, 7 (100%); methotrexate, 5 (71%); mycophenolate, 0; prednisone, 4 (57%)
GVHD	Yes, 6 (40%); no, 9 (60%); median day of onset: 44.5	Yes, 3 (38%); no, 5 (63%) Gut stage 1, skin stage 3 Gut stage 1 Gut stage 1 Median day of onset: 40	Yes, 3 (43%); no, 4 (57%) Gut stage 4 Gut stage 3, skin stage 1 Liver stage 4, skin stage 1 Median day of onset: 52
Antibiotic usage	Levofloxacin prophylaxis, 8 (53%) No antibiotics for neutropenic fever, 1 (7%) Piperacillin/tazobactam or Cefepime first-line, 13 (87%) Piperacillin/tazobactam first-line, 12 (80%) Cefepime first-line, 1 (7%) Other first-line therapy, 1 (7%) Vancomycin added, 7 (47%) Clindamycin added, 4 (27%) Meropenem added, 5 (33%) Clindamycin total usage, 5 (33%) Meropenem total usage, 6 (40%) Metronidazole total usage, 3 (20%) Vancomycin total usage, 7 (47%)	Levofloxacin prophylaxis, 8 (100%) No antibiotics for neutropenic fever, 1 (13%) Piperacillin/tazobactam or Cefepime first-line, 7 (88%) Piperacillin/tazobactam first-line, 6 (75%) Cefepime first-line, 1 (13%) Other first-line therapy, 0 Vancomycin added, 2 (25%) Clindamycin added, 2 (25%) Meropenem added, 0 Clindamycin total usage, 3 (38%) Meropenem total usage, 0 Metronidazole total usage, 1 (13%) Vancomycin total usage, 2 (25%)	Levofloxacin prophylaxis, 0 No antibiotics for neutropenic fever, 0 Piperacillin/Tazobactam or Cefepime first-line, 6 (86%) Piperacillin/tazobactam first-line, 6 (86%) Cefepime first-line, 0 Other first-line therapy, 1 (14%) Vancomycin added, 5 (71%) Clindamycin added, 2 (29%) Meropenem added, 5 (71%) Clindamycin total usage, 2 (29%) Meropenem total usage, 6 (86%) Metronidazole total usage, 2 (29%) Vancomycin total usage, 5 (71%)

SCA indicates sickle cell anemia; AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; HLH, hemophagocytic lymphohistiocytosis; DBA, Diamond Blackfan anemia; FA, Fanconi anemia; SAA, severe aplastic anemia; SCID, severe combined immunodeficiency; SIM, standard-intensity myeloablative; RIM, reduced-intensity myeloablative; NM, nonmyeloablative.

Acute GVHD was staged and graded according to the Glucksberg system [16]. Four patients developed biopsy-confirmed intestinal GVHD (lower gastrointestinal tract only). Six patients met our inclusion criteria for the non-GVHD cohort: absence of any GVHD in any target organ through day 100 after SCT and with survival to at least day +100. Of note, many pediatric SCT programs provide transplantation to adult (>18 years) patients, particularly if the patient was originally diagnosed with the underlying condition as a child. To restrict the analysis to pediatric (<18 years) patients, we removed 2 adult patients from the subsequent analyses, leaving 8 patients (3 with GVHD, 5 without GVHD).

Blinded fecal samples from 10 patients who underwent alloSCT at Cincinnati Children's Hospital were obtained as a validation cohort for the bacterial qPCR experiments. All subjects in the Cincinnati cohort were enrolled in an institutional review board–approved prospective SCT tissue repository (stool, blood, and urine) study. The same inclusion, exclusion, and GVHD categorization criteria used for the Dallas cohort were applied to the Cincinnati cohort. As with the Dallas cohort, 3 adult patients were removed from subsequent analyses, leaving 7 patients (3 with GVHD, 4 without GVHD).

Sample Collection and Isolation of Fecal gDNA

Fresh fecal samples were collected when patients were admitted for SCT and stored at –80°C for further analysis. Subsequent samples were collected roughly every 7 days until stem cell engraftment or resolution of GVHD. gDNA was isolated from fecal specimens as previously described [17]. Briefly, fecal specimens were weighed and suspended in extraction buffer (200 mM NaCl, 200 mM tris(hydroxymethyl)aminomethane, 20 mM EDTA, 6% sodium dodecyl sulfate) and .5 mL of phenol/chloroform/isoamyl alcohol, pH 7.9 (Life Technologies, Carslbad, CA). Cells were lysed by bead beating (.1 mm zirconia/silica beads for bacterial gDNA [BioSpec, Bartlesville, OK]) and subjected to additional phenol/chloroform extractions. Crude DNA extracts were treated with RNase A (Qiagen, Hilden, Germany) and column-purified (PCR Purification Kit, Qiagen). DNA concentrations were quantified by fluorescence-based assay (Quant-iT PicoGreen dsDNA, Life Technologies).

qPCR for Microbiota Analysis

The abundance of the bacterial groups CLEPT (*Clostridium leptum* group; clostridial phylogenetic cluster IV, which measures the Ruminococcaceae family), EREC (*Eubacterium rectale*/*C. coccoides* group; clostridial phylogenic cluster XIVa; which measures the Lachnospiraceae family), and ENTERO (Enterobacteriaceae) were quantified by qPCR analysis (SsoAdvanced SYBR Green Supermix, Bio-Rad, Hercules, CA) using group-specific 16S rRNA gene primers, as previously described [17]. Bacterial abundance was determined using standard curves constructed with reference to cloned DNA corresponding to a short segment of the 16s rRNA gene that was amplified using conserved specific primers. Note that qPCR measures gene copies/g tissue, not actual bacterial numbers or colony-forming units.

16S rRNA Gene Sequencing and Analysis

16S rRNA genes (variable region 4, V4) were amplified from Dallas cohort samples using a composite forward primer and a reverse primer containing a unique 10-base barcode that was used to tag PCR products from respective samples, as previously described [17]. The pooled products were sequenced using the Roche 454 Titanium (Roche, Basel, Switzerland) platform (University of Texas at Austin Genome Sequencing and Analysis Facility) using Roche/454 Titanium chemistry. 16S rRNA sequencing for P10 was performed amplifying the V4 region of the 16S rRNA gene using the same primers as described above but adapted for the Illumina (Illumina, San Diego, CA) platform (Illumina HiSeq 2000, University of Texas Southwestern Medical Center Genomics Core Facility, pair-end sequenced 100 bp reads).

Sequences were quality filtered and analyzed using the open source software package Quantitative Insights Into Microbial Ecology (http://qiime.org/) [18]. 16S rRNA gene sequences were assigned to operational taxonomic units using UCLUST (http://drive5.com/usearch/manual/uclust_algo.html), with a threshold of 97% pair-wise identity, then classified taxonomically using Greengenes (http://greengenes.secondgenome.com/downloads).

MSS Analysis

MSS data (mean of 80.1 million reads per samples; range, 46,343,950 to 194,613,108 reads) was generated from sequencing fecal gDNA fragments from Dallas ($n = 8$) and Cincinnati ($n = 5$) cohort patients on an Illumina HiSeq 2000 (100 bp pair-end reads) in the University of Texas Southwestern Medical Center Genomics Core Facility. Samples collected at the time closest to GVHD diagnosis were used; in non-GVHD patients, the last collected sample was used.

Taxonomic and functional analyses of MSS data were performed as previously described [6]. Briefly, taxonomic composition was performed by using the computational tool MetaPhlAn [19]. Functional pathway abundance was calculated using HUMAnN [20] and FMAP [21]. The open source software package Quantitative Insights Into Microbial Ecology [18] was used to measure the diversity indexes using the species-level MetaPhlAn profiles as input.

For antibiotic resistance gene metagenomic analysis, de novo metagenome assembly and ab initio gene prediction were performed using SPAdes version 3.8.1 [22] and GeneMarkS version 4.32 [23]. MSS reads were then mapped onto the metagenome assembly using BWA version 0.7.12 [24], and read depths for all the predicted genes were estimated using SAMtools version 0.1.19 [25]. All relevant bacterial reference genome sequences (based on the MetaPhlAn results) were downloaded from ftp.ncbi.nlm.nih.gov/genomes. The nucleotide sequences of predicted genes were mapped onto the bacterial reference genome sequences by BWA to identify which specific bacterial genera contain the predicted genes. The protein sequences of all the predicted genes were mapped onto the protein sequences of the antibiotic resistance genes deposited in Antibiotic Resistance Genes Database [26] (23,137 protein sequences, 380 antibiotic resistance gene types, http://ardb.cbcb.umd.edu) by USEARCH version 8.1 (http://drive5.com/usearch).

Murine Model of GVHD

The University of Texas Southwestern Medical Center's institutional animal care and use committee approved all animal experiments performed in this study. Littermates remained cohoused in the same cage to ensure a shared microbiota.

Mice (BALB/c [$H2^d$], Jackson, female, 12 weeks of age) were treated with oral levofloxacin 1 mg/mL in the drinking water 7 days before SCT. BALB/c ($H2^d$) mice were lethally irradiated and underwent transplantation with an intravenous injection of 2×10^7 bone marrow cells (T cell depleted) and 5×10^6 splenic T cells from C57BL/6 ($H2^b$) donors (Jackson, female, 12 weeks of age) [27]. For antibiotic experiments, mice were continued on levofloxacin or levofloxacin/clindamycin (.5 mg/mL) in the drinking water after SCT and for the duration of the experiment. For bacterial supplementation experiments, mice were treated with oral levofloxacin until day +7 (after SCT), levofloxacin/clindamycin from day +8 to day +10 to deplete commensal anaerobes, and then transitioned back to levofloxacin for the remainder of the experiment. Starting on day +11, mice were orally gavaged with a bacterial cocktail consisting of *C. bolteae* (human, American Type Culture Collection [ATCC, Manassas, VA], BAA-613, strain WAL 16351), *Ruminococcus gnavus* (human, ATCC 29149, strain VPI C7-9), *R. torques* (human, ATCC 27756, strain VPI B2-51), and *Blautia producta* (human, ATCC 27340, strain VPI 4299) (1×10^8 colony-forming unit of each strain in a total of .2 mL of sterile PBS) or sterile PBS (.2 mL) every 3 days for the remainder of the experiment. Mice were monitored for up to 60 days for overall survival, daily weights, and clinical GVHD scores [28]. Mice with severe weight loss (>20%) or morbidity were euthanized. All euthanized or deceased mice had spleens and livers harvested and tested for presence of microbial dissemination (organ homogenates were plated on selective media). No mice had evidence of microbial dissemination at time of death. Mouse experiment weight and GVHD scores were analyzed using fitted slopes for individual mice and unpaired *t*-test. Survival was monitored using Kaplan-Meier curves and analyzed with log-rank test. Four mice were used for each group for each experiment. Experiments were performed twice for a total sample size of 8.

Data and Materials Availability

The 16S and MSS data for this study has been deposited in the NCBI Sequence Read Archive: http://www.ncbi.nlm.nih.gov/sra/SRP058320.

Statistical Analyses

Comparison of alpha diversity metrics, bacterial 16S rRNA gene copy numbers, and antibiotic load were analyzed by Mann-Whitney tests, and when multiple comparisons or more than 2 groups were analyzed, Bonferroni's correction to the significance level α was invoked. Linear mixed-effect model analysis was performed in lme4 package (version 1.1-7) and R (version 2.1.2) and the likelihood ratio test was used to assess statistical significance. Clinical correlate analysis was analyzed by chi-square test. Antibiotic load comparisons were analyzed by Mann-Whitney tests. Analysis of whether antibiotic load was associated with GVHD was performed using Firth logistic regression model analysis using logistf package (version 1.21) and R (version 2.1.2). Statistical analyses were carried out using the GraphPad Prism Software (San Diego, CA) unless otherwise indicated.

RESULTS

Declining Levels of AIC Are Associated with the Development of GVHD in Pediatric SCT Patients

We first asked whether changes in specific gut bacterial levels over time were associated with the development of GVHD in pediatric SCT patients. Since 16S sequencing and MSS analyses provide relative abundance data, we utilized bacterial group-

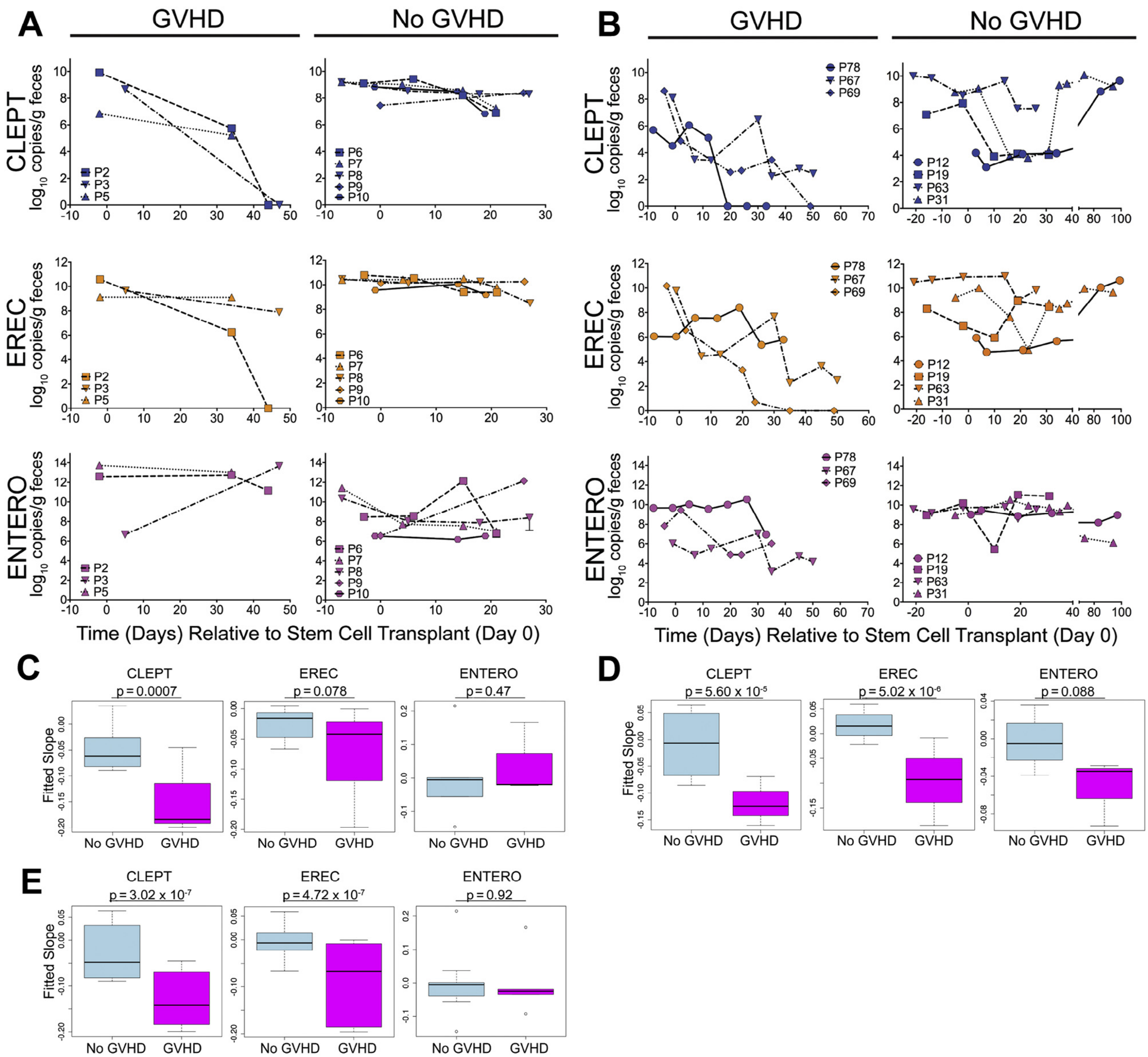

Figure 1. Declining anti-inflammatory Clostridia (EREC and CLEPT) levels are associated with the development of GVHD in pediatric SCT patients. Bacterial group qPCR ($\log_{10}$ copies/g feces) performed on gDNA isolated from fecal specimens collected from patients undergoing allogeneic stem cell transplantation at (A) Children's Medical Center Dallas (n = 8) and (B) Cincinnati Children's Hospital (n = 7). Points represent results from individual patients. Assays were performed in triplicate. ENTERO: Phylum Proteobacteria, Family Enterobacteriaceae. EREC: *Clostridium coccoides-Eubacterium rectale*, Phylum Firmicutes, Clostridial Phylogenetic Cluster XIVa. CLEPT: *Clostridium leptum*, Phylum Firmicutes, Clostridial Phylogenetic Cluster IV. Change in CLEPT, EREC, and ENTERO levels was determined by using fitted slopes. A mixed-linear effect analysis of fitted slopes of CLEPT, EREC, and ENTERO of pediatric SCT patients ± GVHD was performed for the (C) Dallas cohort, (D) Cincinnati cohort, and (E) combined Dallas/Cincinnati cohort. Statistical analysis by likelihood ratio test.

specific qPCR to assess absolute quantitative levels of 3 bacterial groups in serial stool samples from patients undergoing SCT: 2 groups of AIC bacteria, CLEPT (*C. leptum* group), and EREC (*E. rectale/C. coccoides*); and 1 proinflammatory group of bacteria, ENTERO (Enterobacteriaceae family, bacteria associated with IBD and GVHD [29] in mice).

We initially analyzed 24 fecal samples collected from 8 patients who underwent alloSCT at Children's Medical Center Dallas, 3 of whom developed biopsy-confirmed intestinal GVHD (gut stage 1, gut stage 1, and gut stage1/skin stage 3). For GVHD patients, we measured levels bacterial qPCR levels from the first collected stool sample and up to the last collected stool sample closest to the day of GVHD diagnosis. For non-GVHD patients, we measured bacterial levels on all stool samples collected during the inpatient stay. CLEPT and EREC levels decreased (up to 10-log fold in some patients) preceding GVHD diagnosis, whereas in non-GVHD patients, levels were relatively stable (at most a 2-log fold decrease) (Figure 1A). ENTERO levels generally increased in GVHD patients but were more variable in non-GVHD patients. Thus, GVHD patients developed a distinct CLEPT ($P = .00071$, linear mixed-effect model, likelihood ratio test) and EREC ($P = .078$) temporal profile (fitted negative slope) than non-GVHD patients (Figure 1C).

Because previous studies in adult SCT patients have used 16S rRNA sequencing [5,11,30], we also analyzed these same fecal samples using 16S rRNA sequencing and found results consistent with prior published reports—namely, SCT patients with GVHD have a notable decrease in relative abundance of Clostridiales or AIC bacteria (Figure S1).

To validate our bacterial qPCR findings in an independent cohort, we tested 50 additional fecal samples from 7 patients who underwent alloSCT at Cincinnati Children's Hospital. The Cincinnati cohort included 2 patients who developed biopsy-confirmed intestinal (gut stage 4 and gut stage 3/skin stage 1) and 1 with skin (stage 1) and liver (stage 3) GVHD. The non-GVHD group was defined as in the Dallas cohort. Patient number 31 was given a presumptive diagnosis of stage 1 skin GVHD only (no biopsy) and treated with topical steroids only, so patient number 31 was also included in the non-GVHD group. Indeed, consistent with our original analysis, CLEPT and EREC levels declined before the development of GVHD (up to 10-log fold decrease in some patients) but remained more stable (at most, a 3-log-fold decrease) or, in some cases, increased in non-GVHD patients (Figure 1B). Cincinnati GVHD patients, however, had a decrease in ENTERO compared to non-GVHD counterparts (Figure 1B). Again, over the course of SCT, GVHD patients developed a significantly distinct CLEPT ($P = 5.60 \times 10^{-5}$, linear mixed-effect model, likelihood ratio test) and EREC ($P = 5.60 \times 10^{-6}$) temporal profile (negative slope) from their non-GVHD counterparts (Figure 1D).

When applying the linear mixed-effect model analysis to the combined Dallas and Cincinnati cohort data, we found that declining levels of both CLEPT and EREC were significantly associated with GVHD, whereas changes in ENTERO were not (Figure 1E).

Antianaerobic Antibiotic Load Is Associated with GVHD in Pediatric SCT Patients

We then reviewed clinical data of both cohorts to determine if specific clinical characteristics were associated with the development of GVHD. Clinical characteristics of the combined and individual cohorts are shown in Table 1. Briefly, 53% of patients were female with a median age of 10.0 years (range, 1.6 to 14.5). Acute leukemia was the most common diagnosis (acute myelogenous leukemia, 20%; acute lymphoblastic leukemia, 13%). Bone marrow was the source of stem cells in nearly all cases (93%). Conditioning was predominantly myeloablative (standard intensity, 60%; reduced intensity, 40%). Forty-seven percent of transplantations were performed from HLA-matched related donors, 27% from HLA-matched unrelated donors, and 27% from mismatched unrelated donors. GVHD prophylaxis included tacrolimus (53%), cyclosporine (47%), methotrexate (40%), mycophenolate (53%), and prednisone (27%). The incidence of GVHD was 40%, which is consistent with other published reports on pediatric SCT patients [31-34]. Median day of onset of GVHD was day +44.5. Sex, mean age, stem cell source, malignancy diagnosis, and immune prophylaxis therapy were not significantly associated with the development of GVHD (negative data not shown).

All patients received antibiotics during transplantation. Levofloxacin prophylaxis was used in the Dallas cohort only. Piperacillin/tazobactam was the first-line antibiotic for empiric fever and neutropenia in nearly all cases (87%). Use of other antibiotics was more variable and often institution dependent. For instance, clindamycin was used slightly more often and for longer durations in Dallas patients compared with in Cincinnati patients. The rationale for increased clindamycin use for Dallas patients was to provide expanded gram-positive coverage in the setting of persistent fevers and/or other concerning clinical indications (low blood pressures, toxic appearance), and to avoid prolonged use of vancomycin for fear of antibiotic resistance. In 1 case, clindamycin was used because of a documented vancomycin allergy (patient number 5). In contrast, meropenem was used in 86% (6 of 7) Cincinnati patients but in none of the Dallas patients. (Detailed patient antibiotic therapy data are shown in Figures S2 [Dallas cohort] and S3 [Cincinnati cohort]).

Because exposure to antibiotics effective against bacterial anaerobes (specifically, piperacillin-tazobactam and imipenem-cilastin) has been associated with increased GVHD-related mortality in adult SCT patients and mice [11], we asked whether the cumulative exposure to antianaerobic antibiotics was associated with the development of GVHD in pediatric SCT patients, with the rationale that greater cumulative exposure to antianaerobic antibiotics would more effectively deplete anaerobes such as AIC. Total antibiotic [35] and antianaerobic antibiotic load was calculated as the sum of the number of days a patient received any antibiotic or an antianaerobic antibiotic (here defined as clindamycin, meropenem, metronidazole, or piperacillin/tazobactam), respectively. Antibiotic load was calculated up to the day of diagnosis of GVHD or for non-GVHD patients up to day +43, the mean day of GVHD diagnosis for patients who developed GVHD. Both total antibiotic load ($P = .0108$, Mann Whitney) and antianaerobic antibiotic load ($P = .0016$) were significantly higher in GVHD patients than in non-GVHD patients (Figure 2A,B). More importantly, cumulative total antibiotic load ($P = .008$; beta .108 + .055; odds ratio, 1.114; 95% confidence interval, 1.023 to 1.303, Firth logistic regression analysis [36]) and antianaerobic load ($P = .003$; beta .215 + .0120; odds ratio, 1.240; 95% confidence interval, 1.045 to 1.959, Firth logistic regression analysis) was significantly associated with the development of GVHD.

We then asked whether there were differences in specific antibiotic load in patients with or without GVHD. Although meropenem, metronidazole, or piperacillin/tazobactam individual loads were all higher in GVHD patients, only clindamycin load was significantly higher (Figure 2C) ($P = .036$, Mann Whitney) in GVHD patients. Of note, other specific antibiotic loads were not significantly higher in GVHD patients (Figure S4).

In summary, cumulative antibiotic exposure was associated with GVHD. Patients who developed GVHD had significantly higher antibiotic loads, and particularly antianaerobic antibiotic loads, compared with SCT patients without GVHD. Clindamycin load was the only individual antibiotic that was significantly higher in GVHD patients. Overall, increased cumulative exposure to antianaerobic antibiotics was associated with the development of GVHD.

MSS Identifies AIC Species that Are Depleted in GVHD Patients

We performed MSS on select fecal gDNA samples (the samples collected closest to the day of GVHD diagnosis for GVHD patients and last collected sample for non-GVHD patients) to interrogate the genomes of all bacteria residing within any individual patient's gut. Our goal was to gain greater insight into (1) gut microbiota taxonomic composition, particularly at species level resolution; (2) functional pathways, notably the metabolic potential of the microbiome; and (3) the resistome, the collection of all the antibiotic resistance genes in both pathogenic and nonpathogenic bacteria [37].

From a taxonomic standpoint, GVHD patient microbiomes were significantly enriched with *Enterococcus spp.* ($P = .028$ linear discriminant analysis coupled with effect size measurements [38], Kruskal-Wallis test), a finding consistent with

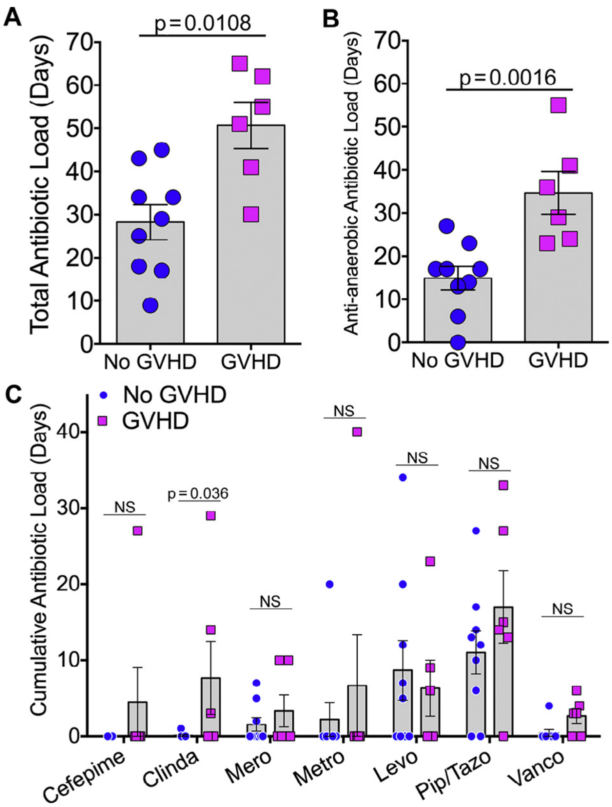

Figure 2. Pediatric SCT patients with GVHD have increased total and antianaerobic antibiotic load. Antibiotic load was calculated as the sum of the number of days an SCT patient received any antibiotic or total load (A), antianaerobic antibiotic (B) (here defined as clindamycin, meropenem, metronidazole, or piperacillin/tazobactam), or (C) specific individual antibiotics. Points represent results from individual patients and horizontal lines with bars representing the mean ± SEM. Experimental differences were analyzed by Mann-Whitney test. Clinda, clindamycin. Mero, meropenem. Metro, metronidazole. Levo, levofloxacin. Pip/Tazo, piperacillin/tazobactam. Vanco, vancomycin.

adult SCT patients with GVHD [5]; the proinflammatory gram-negative family Enterobacteriaceae ($P = .021$), which has been associated with IBD [39] in humans and GVHD in mice [29]; and the aerobic gram-negative family Neisseriaceae ($P = .027$) (Figure 3A,B). In contrast, non-GVHD patient microbiomes were enriched with AIC ($P = .023$ to .039) and 2 other groups of bacteria that have been shown to be beneficial to human health, Bacteroidetes ($P = .044$) and Actinobacteria (*Bifidobacterium spp.*) ($P = .031$) (Figure 3A,B).

Since species-specific effects have been shown to be important (eg, bacteria belonging to the same genus can have significantly different phenotypes or effects on the host [17,40]), we wanted to identify specific species that were enriched or depleted in GVHD or non-GVHD microbiomes. Indeed, our MSS analysis identified specific commensal bacterial species that were abundant in non-GVHD patients but significantly depleted in GVHD microbiomes, 3 of which were AIC: *R. gnavus* ($P = .037$), *R. torques* ($P = .023$), and *Blautia* unclassified ($P = .039$) (Figures 3C and S5A,B). Given the differences in institutional antibiotic practice, we performed additional subanalyses of just Dallas or just Cincinnati patient MSS data, we were able to identify more commensal bacterial species (including another AIC, *C. bolteae*, $P = .025$) that were abundant in non-GVHD Dallas patients but significantly depleted in GVHD microbiomes (Figure S5).

MSS analysis also revealed differences in microbiome gene content (by identifying the presence/absence and calculating abundance of microbial functional pathways) between GVHD and non-GVHD microbiomes. GVHD microbiomes were enriched with bacterial enzymes involved in lipopolysaccharide (LPS) synthesis and comparatively depleted in enzymes involved in core metabolic functions such as carbohydrate, nucleotide, and amino acid metabolism and energy production (Figure S6).

Of note, LPS has been shown to induce inflammation in a number of disease processes, including GVHD [41]. Additionally, GVHD microbiomes were enriched with bacterial fumarate reductase, an enzyme that converts fumarate to succinate. Interestingly, a recent study showed that succinate acts as an innate immune signaling metabolite, sustaining IL-1β production through hypoxia-inducible factor 1-alpha stabilization [42].

Finally, since antibiotics play a critical role in shaping the gut microbiome, we asked whether the resistome, the collection of all the antibiotic resistance genes in both pathogenic and nonpathogenic bacteria [37], was different in GVHD versus non-GVHD microbiomes. In fact, GVHD resistomes did have significantly increased abundance of specific antibiotic-resistance genes when compared with non-GVHD resistomes, yet the resistomes of GVHD patients did not exclusively cluster together (Figure S7).

In summary, MSS analysis revealed that GVHD patients had an increased relative abundance of *Enterobacteriaceae* and *Enterococcus spp.* whereas those without GVHD were enriched with AIC, Bacteroidetes, and *Bifidobacterium*. Specific commensal bacterial species were identified as significantly depleted in GVHD microbiomes, including 4 AIC species. GVHD microbiomes revealed an enrichment of genes involved in LPS biosynthesis and a deficiency in genes involved in core metabolic pathways.

Clindamycin Treatment Depletes Gut AIC and Exacerbates GVHD in Mice

To further explore the potential causality of the observed association of antibiotics and AIC with GVHD, we used a well-established murine model of allo-HSCT and GVHD. We first asked whether clindamycin therapy exacerbates GVHD. BALB/c recipient mice were treated prophylactically with levofloxacin, a practice that is currently used at our institution and others [43]. After transplantation, mice continued to receive either levofloxacin or levofloxacin/clindamycin. Clindamycin-treated mice had significantly decreased survival ($P = .0067$, log-rank test) compared with those mice treated with levofloxacin alone (Figure 4A). Levofloxacin had no significant effect on murine gut AIC levels (as represented by CLEPT qPCR), whereas clindamycin significantly depleted gut AIC ($P = .0167$, Mann-Whitney) (Figure 4B,C).

We then asked whether AIC supplementation after clindamycin therapy could reverse this effect. All recipient mice were initially treated with levofloxacin/clindamycin to deplete gut AIC and then transitioned to levofloxacin alone. Mice were then treated with either an AIC probiotic cocktail (using the 4 AIC species that were most significantly enriched in non-GVHD patients) or a sterile PBS control every 3 days for the remainder of the experiment. AIC-treated mice had significantly longer survival times compared with PBS-treated controls (Figure 4D). AIC supplementation significantly increased AIC levels in the distal gut ($P = .0286$, Mann-Whitney) whereas PBS treatment did not (Figure 4E).

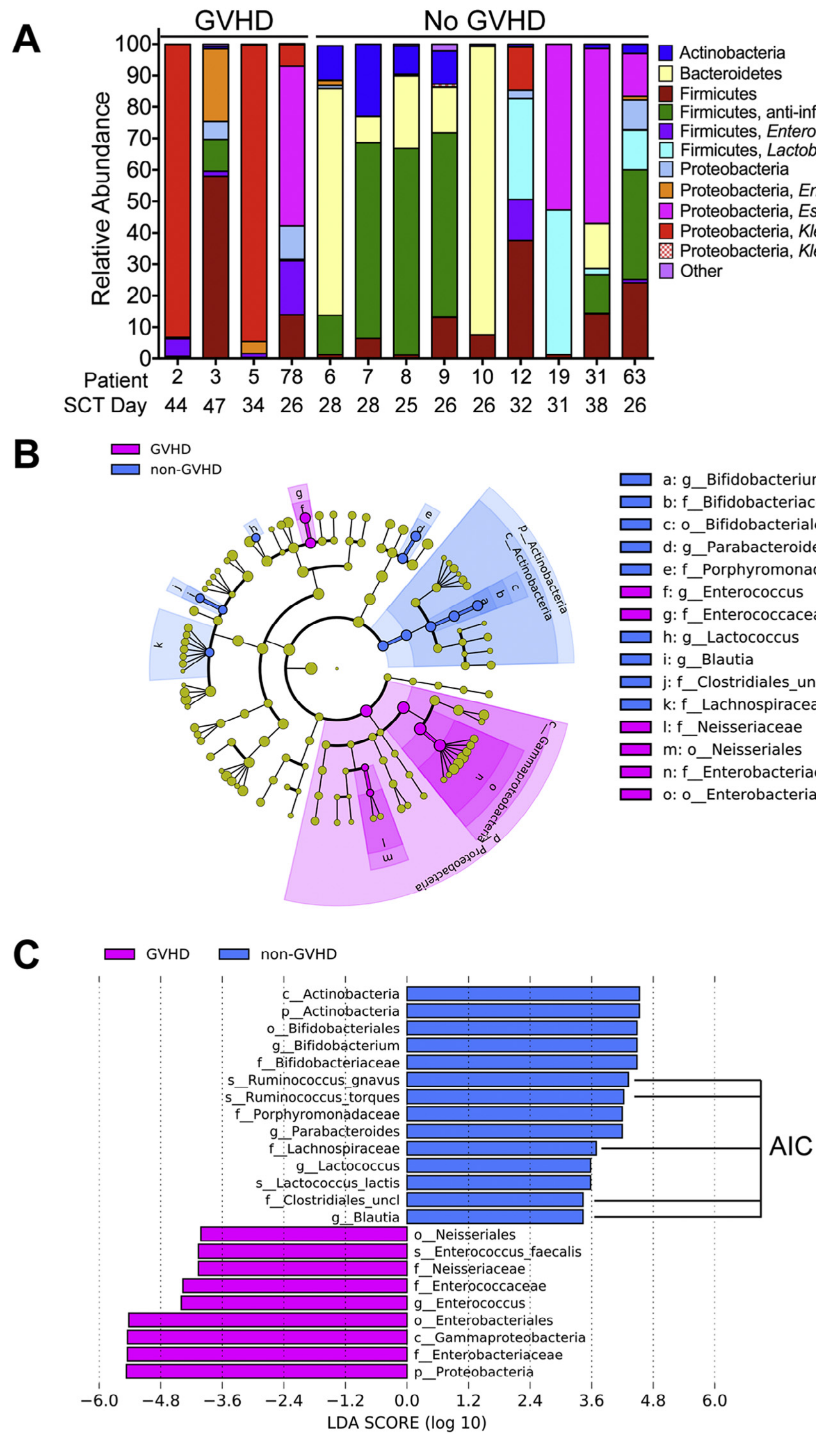

Figure 3. Metagenomic shotgun sequencing (MSS) identifies anti-inflammatory Clostridia bacterial species that are significantly depleted in SCT patients with GVHD. (A) Relative abundance of gut bacterial taxa as determined by MetaPhlAn analysis of MSS data generated from select fecal specimens collected from pediatric SCT patients with and without GVHD. (B,C) Differential taxonomic abundance between GVHD and non-GVHD patients was analyzed by linear discriminate analysis coupled with effect size measurements (LEfSe) projected as a (B) cladogram and (C) histogram. All listed bacterial groups (class, order, family, genus, or species) were significantly ($P < .05$, Kruskal-Wallis test) enriched for their respective groups (GVDH versus non-GVHD). (p), Phylum. (c), Class. (o), Order. (f), Family. (g), Genus. (s), Species. AIC, anti-inflammatory Clostridia.

DISCUSSION

In this study, we observed a significant depletion of AIC preceding the development of GVHD in pediatric SCT patients. The development of GVHD was strongly associated with cumulative antibiotic exposure, particularly antibiotics effective against anaerobes. MSS analysis identified specific commensal bacterial species (including 4 AIC species) that were deficient in GVHD patients. Our clinical observations and associations were validated in a murine model of GVHD. Clindamycin-depleted gut AIC exacerbated GVHD, and oral supplementation of AIC increased AIC levels and attenuated GVHD.

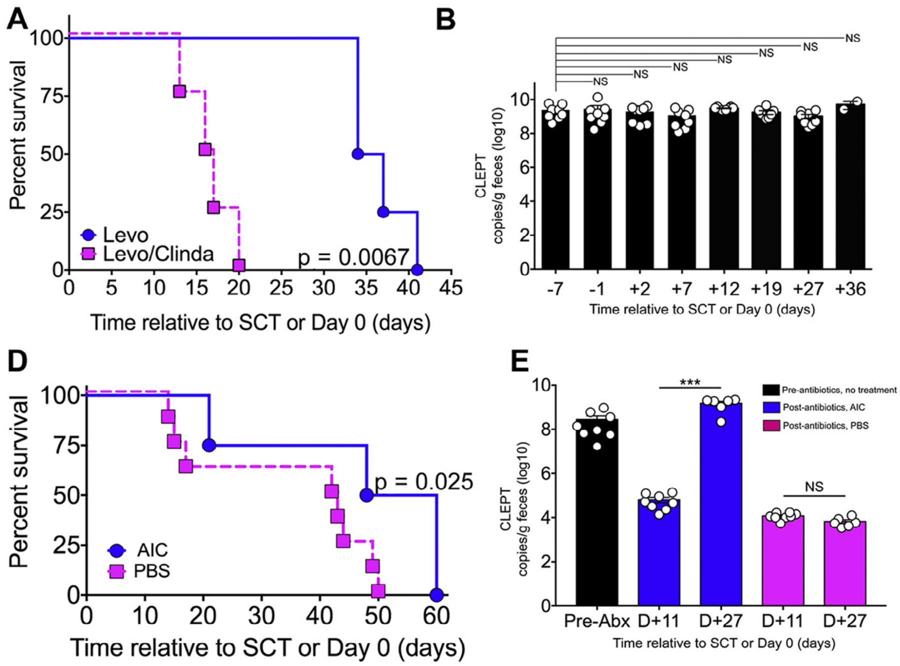

Figure 4. Clindamycin treatment depletes anti-inflammatory Clostridia (AIC) and exacerbates GVHD whereas AIC supplementation ameliorates GVHD in mice. Lethally irradiated Balb/C (H2^d) recipients were underwent transplantation with 2×10^7 C57BL/6 (H2^b) T cell–depleted bone marrow cells and 5×10^6 splenic T cells. Recipients were initially treated with levofloxacin. After transplantation (day 0), mice were treated with levofloxacin or levofloxacin/clindamycin in the drinking water for the duration of the experiment. For AIC supplementation experiments, recipients were initially treated with levofloxacin until day +7, then levo/clindamycin until day +10, then resumed levofloxacin until completion of the experiment. Starting on day +11 mice were orally gavaged with an AIC probiotic cocktail (1×10^8 colony-forming unit of *Clostridium bolteae*, *Ruminococcus gnavus*, *R. torques*, and *Blautia producta* in .2 mL sterile PBS) or sterile PBS (.2 mL) every 3 days for the remainder of the experiment. Survival curves of mice that had undergone transplantation and were receiving (A) levofloxacin versus levofloxacin/clindamycin and (D) AIC versus PBS. n = 8 per group. Statistical analysis by log-rank test. AIC levels as determined by CLEPT qPCR performed on fecal gDNA derived from stool specimens collected throughout transplantation for mice receiving (B) levofloxacin, (C) levofloxacin/clindamycin, and (E) AIC versus PBS treatment. Bars represent the mean + SEM. Points represent results from individual animals. Statistical analysis by Mann-Whitney test. * $P < .05$. ** $P < .01$. *** $P < .001$. NS, not significant.

The gut microbiota changes in pediatric SCT patients with GVHD are consistent with previous studies in adult SCT patients [4,5,30]. Depletion of commensal anaerobes (particularly *Clostridiales* or what we term AIC) appears to occur in both adult [4,11,30] and pediatric patients who develop GVHD. Expansion of *Enterococcus spp.* also appears to accompany AIC depletion in both adults [5] and children. Interestingly, an increased relative abundance of the proinflammatory gram-negative bacteria (Enterobacteriaceae) was noted in our MSS analysis, but a quantitative increase in ENTERO qPCR was not noted. We have reported similar findings previously [17]: qPCR data can show a significant (5 to 10 log-fold) decrease in anaerobic bacteria such as AIC (which accounts for >95% of the gut microbiota) accompanied by no significant change in ENTERO levels, while sequencing data shows an increase in ENTERO relative abundance. Thus, the apparent "expansion" of ENTERO noted in the sequencing data analysis is really due to the significant drop in absolute AIC levels. MSS analysis also identified other potentially beneficial bacterial species enriched in non-GVHD patients. Interestingly, while Dallas non-GVHD patients were enriched with AIC and Bacteroidetes, Cincinnati non-GVHD had significantly more *Lactobacillus* species (Figure S5). *Lactobacillus* species have been reported as prominent members of the gut microbiome in large studies of healthy children [44,45]. Of note, *L. rhamnosus* (1 of the *Lactobacillus* species identified to be abundant in Cincinnati non-GVHD patients) has been used as a probiotic to treat childhood diarrheal illness [46]. Finally, *Lactobacillus* has been shown to attenuate GVHD in mice [4]. The taxonomic differences between Dallas and Cincinnati non-GVHD patients are most likely attributable to institutional differences in antibiotic therapy. But these data suggest that different gut microbiota communities (eg, AIC/Bacteroidetes abundant versus *Lactobacillus* abundant) may afford protection against GVHD.

One limitation of our study was that many of the patients enrolled in the Dallas cohort had to be excluded from the study because of insufficient amounts of DNA required to complete all 3 gut microbiota profiling methods. This was possibly a result of the use of broad-spectrum antibiotics in many patients, resulting in fewer overall gut microbiota and, thus, lower concentrations of fecal gDNA. Thus, our study may have had a bias towards inclusion of patients who did not receive antibiotics and those who received antibiotics but nevertheless had amplifiable DNA because they harbored antibiotic-resistant gut microbiota. These are important factors to consider in future gut microbiome studies involving patients receiving broad-spectrum antibiotics.

It is still unclear whether specific gut microbiota are causative or merely associated with GVHD. AIC have been shown to induce regulatory T cells and IL-10 and dampen inflammation in IBD in mice [7,8]. In fact, short-chain fatty acids produced by AIC (in the absence of bacteria) can mediate these effects [47]. A recent study showed that a probiotic cocktail of 17 AIC species mitigates GVHD in mice and the short-chain fatty acid butyrate restores intestinal epithelial cell junctional integrity and decreases apoptosis [9]. In this study, we used 4 AIC (2 of which belong to the 17 AIC cocktail that has been used to attenuate IBD [7] and GVHD [9] in mice) to decrease GVHD severity in mice. The obvious question is whether use of probiotics is a safe and viable therapy for GVHD patients. The first pilot study using fecal microbiota transplantation to treat 4 patients with steroid-refractory GVHD found the probiotic to be both safe and effective [48]. However, there is still a theoretical concern about introducing bacteria into an immunocompromised host, so further studies will need to be done.

Because early diagnosis and treatment of GVHD are associated with improved outcomes [2], developing reliable noninvasive GVHD biomarkers is crucial. miRNAs [49], immune cells (particularly regulatory T cells) [50-52], and proteins, such as Reg3α [53], have shown promise as noninvasive biomarkers for GVHD, but there is no clinical consensus on their use. The idea of monitoring changes in the gut microbiome as a noninvasive biomarker for GVHD is intriguing. Unfortunately, rapid and cost-effective gut microbiota next-generation sequencing (16S rRNA and MSS) is currently not feasible for clinical use. Thus, the use of qPCR using primers specific for key bacterial groups is an attractive alternative. Although the qPCR results from this study are encouraging, further studies with a larger sample size are needed to confirm our findings and to validate whether a clinical laboratory can feasibly employ this qPCR assay in real time. Of note, a recent study showed that a gut microbiota-derived metabolite measured in the urine correlated with the abundance of AIC and showed promise as a biomarker assay for GVHD [54].

Although probiotic therapy and gut microbiota monitoring are not yet ready to be incorporated into the clinical care of SCT patients, changes to the current use of antibiotic therapy in SCT patients should be considered. Antibiotics, particularly antianaerobic antibiotics [30], appear to be critical drivers of the gut microbiota changes associated with GVHD. In this study and others, the use of specific antianaerobic antibiotics (eg, clindamycin, piperacillin-tazobactam [11], imipenem-cilastin [11]) was associated with the development of GVHD in patients and exacerbated GVHD in mice. While these specific antibiotics do provide added protection against anaerobic bacterial infections, infections with anaerobic bacteria are rare [55]. In fact, most peri-transplantation–period infections are caused by gram-negative bacteria (Enterobacteriaceae), gram-positive bacteria (coagulase-negative Staphylococci, *S. viridans*, and *Enterococcus spp.*), and *Candida* species [55]. The question is whether we should tailor the use of antibiotics in SCT (using more narrow-spectrum antibiotics) to maximize protection against systemic microbial infections while minimizing damage to beneficial members of the microbiome, such as AIC. Further studies will need to be done to determine exactly which antibiotics (types, combinations, and cumulative load) should be used and/or avoided during SCT.

Finally, given our focus on pediatric SCT patients, our original intention was to use a pediatric SCT GVHD preclinical model. Unfortunately, we were not able to consistently provide transplantation to pediatric mice (14- to 21-day-old mice) and we observed a high rate of bacterial dissemination despite using antibiotic prophylaxis. Therefore, we were relegated to using a conventional adult mouse GVHD model. In the end, however, using a pediatric model may not have been completely necessary as the 2 youngest patients in our study did not display "immature" gut microbiota (lack of Firmicutes and Bacteroidetes) observed in children < 1 to 2 years old. Nonetheless, further studies will need to be done in these younger children to see whether they are at increased risk for developing GVHD.

In conclusion, there is mounting evidence that the gut microbiota plays a critical role in the regulation of host immune function. Therefore, in a clinical field in which host immune function is constantly in flux, it is not surprising that the gut microbiome likely plays a critical role in the SCT patient's health. In the future, just as we routinely give systemic immunosuppressive agents to prevent or treat GVHD, we may also use novel means to augment gut immune effectors to modulate GVHD.

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SUPPLEMENTARY DATA

Supplementary data related to this article can be found online at doi:10.1016/j.bbmt.2017.02.004.

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